# GCC Database Partner API

Amazon Developer Integration Guide

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> **What's new in v1.1 (April 2026):** Per-identifier product names are now surfaced in the GS1 response. Each `additionalTradeItemId` entry optionally carries a `name` field. The data is already collected during import and on the certificate edit page, and has been available in the Standard endpoint as `productIds[].productName`. This update brings the GS1 endpoint into parity — no data model or UI changes required, simply a serialization update on a field that's already populated. See §5.2 for the response shape and Appendix A for the field mapping.

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## 1. Quick Start

Three steps to get your first data:

1. **Get your API key** from GCC (format: `gcc_partner_[hash]`). This full string, including the `gcc_partner_` prefix, is your `X-API-Key` value — use it exactly as provided in the header.
2. **Make your first request:**

```
# Test environment (current)
curl -X GET "https://gccwebapp.com/api/v1/certificates/gs1" \
  -H "X-API-Key: YOUR_API_KEY" \
  -H "Accept: application/json"

# Production environment (pending DNS configuration)
curl -X GET "https://greencirclecertified.com/database/api/v1/certificates/gs1" \
  -H "X-API-Key: YOUR_API_KEY" \
  -H "Accept: application/json"
```

3 **You'll receive** a JSON response with all active certificates tagged to your partner account, including ASINs, certification details, and PDF download links.

> **Integration model:** You PULL data from GCC. GCC does not push to you. Call the API on your schedule to retrieve current certification data. Each API call returns the active certificate database. Amazon is responsible for identifying what is new or removed on their side to badge and unbadge products accordingly.

## 2. Authentication

Every request requires the `X-API-Key` header:

```
X-API-Key: gcc_partner_fc8498d5e8067766f74ed28bb439b17210196895e4e67828036f1737d222138a
```

| Response Code | Meaning | Action |
|---|---|---|
| `200` | Success | Process the response |
| `401` | Invalid or missing API key | Check your key in the header |
| `403` | Endpoint not permitted | Contact GCC support |
| `404` | Resource not found | Verify certificate number or product ID |
| `429` | Rate limit exceeded | Wait and retry (200 API calls/hour) |
| `500` | Server error | Contact GCC support with request details |

## 3. API Endpoints

| Environment | Base URL | Status |
|---|---|---|
| **Test (current)** | `https://gccwebapp.com/api/v1` | Active — use this for development and integration testing |
| **Production** | `https://greencirclecertified.com/database/api/v1` | Pending DNS configuration |

### GET `/certificates/gs1` RECOMMENDED

List all certificates in GS1 JSON-LD format. ASINs and product IDs are always included.

**Parameters:** `pageNumber` (default: 1), `pageSize` (default/max: 100), `searchTerm`, `sortBy`

### GET `/certificates/gs1/{certNumber}`

Get a single certificate by number in GS1 format.

**Example:** `/certificates/gs1/24-0366`

### GET `/certificates`

List all certificates in standard JSON format.

**Parameters:** `pageNumber`, `pageSize`, `searchTerm`, `sortBy`, `includeProductIds` (boolean - set to `true` to include ASINs)

### GET `/certificates/number/{certNumber}`

Get a single certificate by number in standard JSON format.

### GET `/certificates/search/productid`

Find a certificate by ASIN, UPC, GTIN, or EAN.

**Parameters:** `productId` (required), `type` (ASIN, UPC, GTIN, EAN)

**Example:** `/certificates/search/productid?productId=B08XQRT456&type=ASIN`

**Note:** At a minimum, Amazon will be searching by ASIN.

### GET `/certificates/{certNumber}/download`

Download the certificate PDF document. Returns `application/pdf`.

## 4. Response Data Model

### Certificate Fields

| Field | Type | Description | Example |
|---|---|---|---|
| `certificateNumber` | string | Unique certificate ID | `24-0366` |

| Field | Type | Description | Example |
|---|---|---|---|
| `status` | string | Always "Active" (only active certs returned) | `Active` |
| `issueDate` | datetime | Certificate effective date | `2024-01-15T00:00:00Z` |
| `expirationDate` | datetime | Certificate expiration date | `2025-01-15T00:00:00Z` |
| `company` | string | Certified company/brand name | `EcoProduct Co` |
| `productName` | string | Certified product name | `Recycled Paper Towels` |
| `certificationType` | string | Certification standard | `Sustainable Energy Practices` |
| `amazonCode` | string | Amazon's standardized cert identifier | `greencircle_sustainable_energy_practices` |
| `certificationResults` | string | Results achieved | `25%` |
| `pathway` | string | Certification pathway* | `Energy usage reduction` |
| `issuer` | string | Certifying body | `GreenCircle Certified` |
| `certificationCountries` | string | Marketplace countries** | `All` |
| `pdfDownloadUrl` | string | Relative URL to download cert PDF | `/api/v1/certificates/24-0366/download` |

* Pathway will not be applicable for all certification types and will be left blank if not relevant.
** Marketplace IDs will not be passed. Amazon will apply the badge in all marketplaces where the product is sold, as the certificate is not restricted by location.

> **NOTE:** Only certificates that are currently active will be shared with Amazon. Any certificates that were previously badged but are no longer in the list/active should be unbadged. Amazon will manage the comparison and handle removals on their side.

### Product Identifier Fields (within `productIds[]`)

| Field | Type | Description | Example |
|---|---|---|---|
| `type` | string | ASIN, UPC, GTIN, EAN | `ASIN` |
| `value` | string | The product identifier value | `B08XQRT456` |
| `status` | string | Product ID status | `Active` |
| `partner` | string | Associated partner | `Amazon` |
| `productName` | string | Product variant name | `Paper Towels - 12 Pack` |
| `productSku` | string | Company SKU. **Note:** SKU is not always provided by the certificate holder. ASIN will always be passed at minimum. If SKU is not available, this field will be blank. | `RPT-12PK` |

## 5. Request/Response Examples

### 5.1 List All Certificates (Standard JSON)

#### Request

```
GET /api/v1/certificates?includeProductIds=true
X-API-Key: YOUR_API_KEY
```

#### Response

```
{
  "success": true,
  "pagination": {
    "totalCount": 1,
    "pageNumber": 1,
    "pageSize": 100,
    "totalPages": 1,
    "hasNext": false,
    "hasPrevious": false
  },
  "data": [
    {
      "certificateNumber": "24-0366",
      "status": "Active",
      "issueDate": "2024-01-15T00:00:00Z",
      "expirationDate": "2025-01-15T00:00:00Z",
      "certificationCountries": "All",
      "partner": "Amazon",
      "certificationResults": "32%",
      "pathway": "Energy usage reduction",
      "company": "EcoProduct Co",
      "productName": "Energy-Efficient LED Bulb",
      "certificationType": "Sustainable Energy Practices",
      "amazonCode": "greencircle_sustainable_energy_practices",
      "issuer": "GreenCircle Certified",
      "productIds": [
        {
          "type": "ASIN",
          "value": "B08XQRT456",
          "status": "Active",
          "partner": "Amazon",
          "productName": "Energy-Efficient LED Bulb",
          "productSku": "LED-BULB-001"   // May be blank if not provided by certificate holder
        }
      ],
      "pdfDownloadUrl": "/api/v1/certificates/24-0366/download"
    }
  ]
}
```

### 5.2 List All Certificates (GS1 Format)

> **Note:** GS1 JSON-LD uses field names defined by the GS1 Web Vocabulary specification, which differ from the Standard JSON field names. For example, `productName` becomes `name` and `certificationType` becomes `program`. See Appendix A for the complete field mapping and an explanation of why the names differ.

#### Request

```
GET /api/v1/certificates/gs1
X-API-Key: YOUR_API_KEY
```

#### Response

**Note:** The response follows the GS1 Certification Data Model. Two fields ( `amazonCode` and `pathway` ) are GCC extensions not defined in the GS1 standard — they are included to support Amazon's Climate Pledge Friendly program mapping.

```
{
  "@context": "https://gs1us-technology.github.io/certification-data-model/cert_context.json",
  "$schema": "https://gs1us-technology.github.io/certification-data-model/cert_schema.json",
  "id": "https://greencirclecertified.com/database/api/v1/certificates/gs1/24-0366",
  "type": "CertificationReport",
  "issueDate": "2024-01-15",
  "issuer": "GreenCircle Certified",
  "products": [
    {
      "id": "https://greencirclecertified.com/database/api/v1/certificates/gs1/24-0366/product",
      "type": "Product",
      "gtin": null,                                          // Amazon-tagged certs typically have
no GTIN; ASIN is the primary identifier
      "name": "Energy-Efficient LED Bulb",                   // → productName
      "additionalTradeItemId": [                              // → productIds[]
        { "type": "ASIN", "value": "B08XQRT456", "name": "Energy-Efficient LED Bulb - 4 Pack" }
      ],
      "certificationInfo": {
        "type": "CertificationDetails",
        "agency": "GreenCircle Certified",                    // → issuer
        "agencyURL": "https://www.greencirclecertified.com",
        "program": "Sustainable Energy Practices",            // → certificationType
        "certificateId": "24-0366",                           // → certificateNumber
        "status": "ACTIVE",                                   // → status ("Active")
        "value": "32%",                                       // → certificationResults
        "startDate": "2024-01-15",                            // → issueDate
        "endDate": "2025-01-15",                              // → expirationDate
        "targetCountries": ["All"],                           // → certificationCountries
        "amazonCode": "greencircle_sustainable_energy_practices",  // → amazonCode (GCC extension)
        "pathway": "Energy usage reduction"                   // → pathway (GCC extension)
      }
    }
  ]
}
// See Appendix A for complete GS1 ⇔ Standard JSON field mapping
```

### 5.3 Search by ASIN

#### Request

```
GET /api/v1/certificates/search/productid?productId=B08XQRT789&type=ASIN
X-API-Key: YOUR_API_KEY
```

#### Response

```
{
  "success": true,
  "data": {
    "certificateNumber": "24-0512",
    "status": "Active",
    "certificationResults": "60%",
    "company": "CleanGreen Industries",
    "productName": "Recycled Paper Towels",
    "certificationType": "Recycled Content",
    "amazonCode": "greencircle_recycled_content",
    "productIds": [
      { "type": "ASIN", "value": "B08XQRT456", "productName": "Paper Towels - 6 Pack" },
      { "type": "ASIN", "value": "B08XQRT789", "productName": "Paper Towels - 12 Pack",
"isMatchedId": true },
      { "type": "ASIN", "value": "B08XQRT999", "productName": "Paper Towels - 24 Pack" }
    ],
    "pdfDownloadUrl": "/api/v1/certificates/24-0512/download"
  }
}
```


> **Note:** Searching for one ASIN returns the full certificate with ALL associated product IDs. The `isMatchedId: true` flag marks which one matched your search.

### 5.4 Download Certificate PDF

#### Request

```
GET /api/v1/certificates/24-0366/download
X-API-Key: YOUR_API_KEY
```

Returns the PDF file with `Content-Type: application/pdf` .

## 6. Error Handling

All errors return a consistent JSON structure:


```
{
  "success": false,
  "error": "Description of what went wrong"
}
```


| Code | Error | Resolution |
|---|---|---|
| `401` | Missing or invalid API key | Ensure `X-API-Key` header is present with a valid key |
| `403` | Insufficient permissions | Your key may not have access to this endpoint |

| Code | Error | Resolution |
|---|---|---|
| `404` | Certificate or product ID not found | Verify the certificate number or ASIN exists and is Active |
| `429` | Rate limit exceeded | Default: 200 API calls/hour (each call can return up to 100 certificates). Wait and retry with exponential backoff |
| `500` | Server error | Contact GCC support with the request details |

## 7. Pagination

### How It Works

The API paginates by **certificate**, not by product or ASIN. Each page returns up to **100 certificates per request**, and each certificate includes **all** of its associated ASINs and product identifiers in a single response — product IDs are never split across pages.

> **Example:** If a single certificate covers 50 ASINs, all 50 ASINs are returned with that certificate in one response. A page of 100 certificates could contain thousands of ASINs.

### Capacity

| Parameter | Value |
|---|---|
| Max certificates per page | 100 |
| Partner API rate limit | 200 calls/hour |
| Max certificates per hour | 20,000 (200 calls × 100 per page) |
| ASINs per certificate | Unlimited — all included in the response |

For example, if Amazon has 300 active certificates covering 11,000+ ASINs, that requires only **3 API calls** (3 pages × 100 certificates) to retrieve the complete dataset — well within the 200 calls/hour limit. Rate limits reset at the top of each hour.

### Pagination Example

```
// Page 1
GET /api/v1/certificates?pageNumber=1&pageSize=100

// Check response
"pagination": {
  "totalCount": 250,
  "pageNumber": 1,
  "pageSize": 100,
  "totalPages": 3,
  "hasNext": true,      // ← more pages available
  "hasPrevious": false
}

// Page 2
GET /api/v1/certificates?pageNumber=2&pageSize=100

// Continue until hasNext is false
```

Loop until `hasNext` is `false` to retrieve all certificates.

> **Page size & rate limits are configurable.** The default maximum of 100 certificates per page and 200 API calls per hour are suitable for most integration scenarios. If your volume requires higher throughput, contact GCC support to adjust these limits for your API key.

## 8. Security & Access Scope

| Control | Details |
|---|---|
| **Transport** | HTTPS/TLS required for all requests |
| **Authentication** | API key via `X-API-Key` header |
| **Data Isolation** | Your key only returns certificates where Partner = "Amazon"; no other partner data is accessible |
| **Status Filter** | Only active certificates are returned; draft, expired, suspended, and revoked certs are excluded |
| **Rate Limiting** | 200 API calls/hour (each call returns up to 100 certificates; configurable per key) |
| **Key Management** | Keys issued and revocable by GCC administrators |

> **You will never see:** other partner data, internal notes, workflow history, draft certificates, or GCC user information.

## 9. Field Reference

### Standard JSON vs GS1 JSON-LD Field Mapping

| Data | Standard JSON Path | GS1 JSON-LD Path |
|---|---|---|
| Certificate Number | `data[].certificateNumber` | `products[].certificationInfo.certificateId` |
| Status | `data[].status` ("Active") | `certificationInfo.status` ("ACTIVE") |
| Issue Date | `data[].issueDate` (ISO 8601+time) | `issueDate` (date only) |
| Expiration Date | `data[].expirationDate` | `certificationInfo.endDate` |
| Cert Results | `data[].certificationResults` | `certificationInfo.value` |
| Cert Type | `data[].certificationType` | `certificationInfo.program` |
| Amazon Code | `data[].amazonCode` | `certificationInfo.amazonCode` |
| ASIN | `data[].productIds[].value` | `products[].additionalTradeItemId[].value` |
| GTIN | `data[].productIds[].value` | `products[].gtin` (14-digit padded) |
| PDF Link | `data[].pdfDownloadUrl` | *(use Standard endpoint)* |

### Product Identifier Types

| Type | Format | Example |
|---|---|---|
| ASIN | 10 characters (alphanumeric) | `B08XQRT456` |
| UPC | 12 digits | `012345678901` |
| GTIN | 14 digits | `00012345678901` |
| EAN | 13 digits | `4006381333931` |

**Note:** Only what is available will be passed; not all identifier types will be provided. At a minimum, ASIN will be included.

### Available Sort Fields

| sortBy Value | Description |
|---|---|
| `certificateNumber` | Sort by certificate number |
| `issueDate` | Sort by issue date |
| `expirationDate` | Sort by expiration date |
| `productName` | Sort by product name |
| `company` | Sort by company |

### GCC Support Contact

For API key requests, issues, or questions: Contact GreenCircle Certified.

# Appendix A: GS1 JSON-LD Format Reference

### Why Two Response Formats?

The GCC API provides two response formats for certificate data:

| Format | Endpoint | Purpose |
|---|---|---|
| **Standard JSON** | `/certificates` | GCC's own field naming convention. Simple and direct. |
| **GS1 JSON-LD** | `/certificates/gs1` | Follows the GS1 Certification Data Model specification. Industry-standard format for interoperability. |

Both formats return the same underlying data. The difference is the field names and structure.

### What is GS1?

**GS1** is the global standards organization that governs product identification across the supply chain — including UPC barcodes, GTINs, and EAN codes used by over 2 million companies worldwide. GS1 is a not-for-profit headquartered in Brussels with member organizations in over 110 countries.

**GS1 US**, the American member organization, published the **Certification Data Model** — a standardized format for exchanging product certification information between certification bodies and retailers. This is the specification the `/certificates/gs1` endpoint follows.

### Why Do the GS1 Field Names Differ from Standard JSON?

The GS1 JSON-LD format uses field names defined by the **GS1 Web Vocabulary**, a formal specification maintained by GS1 Global. These field names cannot be changed because:

- The response references a published JSON Schema ( `cert_schema.json` ) and JSON-LD context ( `cert_context.json` ) hosted by GS1 US
- The schema enforces required field names, data types, and validation rules (e.g., `gtin` must be exactly 14 digits, `status` must be `ACTIVE` or `INACTIVE` )
- Any consumer validating against the schema would reject or ignore non-standard field names
- JSON-LD consumers map field names to GS1 vocabulary URIs — renamed fields would lose their semantic meaning

### Complete Field Name Mapping

The following table shows how each piece of data maps between the two formats:

| Data | Standard JSON Field | GS1 JSON-LD Field | GS1 Spec Definition |
|---|---|---|---|
| Product Name | `productName` | `name` | "Consumer friendly short description of the product" |
| Certificate Number | `certificateNumber` | `certificateId` | "A vendor specific identifier for the certificate" |
| Certification Type | `certificationType` | `program` | "The certification name or umbrella initiative" |

| Data | Standard JSON Field | GS1 JSON-LD Field | GS1 Spec Definition |
|---|---|---|---|
| Certification Results | `certificationResults` | `value` | "The certification's standard value or level" |
| Issue Date | `issueDate` (ISO 8601 + time) | `startDate` (date only) | "First date of validity for the certification" |
| Expiration Date | `expirationDate` (ISO 8601 + time) | `endDate` (date only) | "Last date of validity for the certification" |
| Countries | `certificationCountries` | `targetCountries` | "Countries where the certification applies" |
| Status | `status` ("Active") | `status` ("ACTIVE") | GS1 requires uppercase: `ACTIVE` or `INACTIVE` |
| Issuer | `issuer` | `agency` | "The certification agency or issuing body" |
| ASIN / Product IDs | `productIds[]` | `additionalTradeItemId[]` | GS1 standard for non-GTIN identifiers |
| Per-Identifier Product Name | `productIds[].productName` | `additionalTradeItemId[].name` | Optional. Product name associated with this specific identifier (e.g. variant or pack-size name). Omitted when not supplied during import. |
| GTIN | `productIds[].value` | `gtin` | 14-digit padded Global Trade Item Number |
| PDF Link | `pdfDownloadUrl` | *N/A — use Standard endpoint* | Not part of GS1 spec |

### GCC Custom Extensions

The following fields are **GCC-specific extensions** included in the GS1 response for Amazon integration. They are not part of the GS1 specification but are safe to include — the GS1 schema allows additional properties.

| Field | Description | Notes |
|---|---|---|
| `amazonCode` | Amazon's standardized certification identifier | Used by Amazon to map certificates to Climate Pledge Friendly badge categories |
| `pathway` | Certification pathway | Not applicable for all certification types; blank if not relevant |

### Governance & Schema References

| Resource | URL |
|---|---|
| GS1 Web Vocabulary (global standard) | `gs1.org/voc/` |
| GS1 CertificationDetails class | `gs1.org/voc/CertificationDetails` |
| Certification Data Model — JSON-LD Context | `gs1us-technology.github.io/certification-data-model/cert_context.json` |
| Certification Data Model — JSON Schema | `gs1us-technology.github.io/certification-data-model/cert_schema.json` |

> **Recommendation:** For most Amazon integration use cases, the GS1 endpoint ( `/certificates/gs1` ) is recommended. It provides an industry-standard format that ensures interoperability and future compatibility as GS1 certification exchange becomes more widely adopted across the retail supply chain.

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